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Editorial Office
Applied Energy
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Dear Editor,

We are pleased to submit our manuscript entitled **"Spectral Co-Design of Agrivoltaic Modules for Multifunctional Net Energy Benefit: A Quantum-Guided Digital Twin"** for consideration as a Research Article in *Applied Energy*. It is not under consideration elsewhere. The journal's systems-level treatment of energy conversion, the water–energy–food nexus, lifecycle assessment, and deployment economics is the natural home for this work.

Central question and significance. Can spectral co-design guided by non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (power + water + carbon + diagnostics) in a field-constrained agrivoltaic stack, and which barriers remain physical rather than economic? Static semi-transparent modules resolve the food–energy land conflict only by uniformly sacrificing electrical output or crop productivity. We show that an organic photovoltaic layer coupled to a nanoparticle-on-mirror (NPoM) cavity can actively shape the transmitted spectrum, protect vibronic coherence in a model photosynthetic complex, and host label-free SERS diagnostics—while a digital twin maps these device-level choices onto canopy yield, FAO-56 evapotranspiration, Net Ecological Benefit (NEB), and cooperative payback.

Headline results (all modeled, fully traceable).

- Dual-band transmission at 750 nm and 820 nm extends exciton coherence by 50 % and raises forward transfer yield by 25 %.
- Confining the NPoM sentinel to 1 % of canopy area absorbs a 91.8 % local transport penalty while preserving global device-level yield $\Phi = 0.971$ (vs. passive baseline 0.980).
- The smart shield cuts crop evapotranspiration by 28 % ($460\,\mathrm{L\,m^{-2}\,yr^{-1}}$) and delivers a NEB of $72.6\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$.
- For a modeled $500\,\mathrm{m^2}$ cooperative module in Cameroon, payback is 4.32 yr at 30 % subsidy (2.68 yr with local materials).

Why *Applied Energy*? The manuscript integrates photovoltaics, microclimate (FAO-56), lifecycle carbon accounting, cooperative techno-economics, and secure telemetry into a single decision-relevant framework—precisely the multi-scale, applied scope of the journal. Quantum methods appear only as the design engine for the spectral architecture; the value chain (what the quantum layer buys, and what it costs) is stated explicitly for an energy-systems readership.

Novelty and positioning. To our knowledge, this is the first agrivoltaic stack whose spectral architecture is co-designed with numerically exact open-quantum-system dynamics, the first quantification of the local-to-global transport trade-off that makes a plasmonic diagnostic layer viable at canopy scale, and the first multifunctional TEA/LCA that internalizes power, water, carbon, and sensing in a single NEB and payback account. Economic barriers are shown to be tractable; physical barriers (nanofabrication, fouling, encapsulation) are stated honestly.

Compliance. The manuscript has been language-edited; all authors approve the submission and declare no competing interests. An earlier version was considered at *Nature Energy*; we submit here as the more suitable venue. Subscription route (no APC). Corresponding author based in Cameroon.

Suggested reviewers: A. Olaya-Castro (UCL), J. Yuen-Zhou (UCSD), J. J. Baumberg (Cambridge), G. A. Barron-Gafford (Arizona)—covering quantum biology, polaritonic chemistry, NPoM/SERS, and agrivoltaics.

Sincerely,

Steve Cabrel Teguia Kouam (on behalf of all authors)
Enclosures: Manuscript, Supplementary Material, Highlights, Graphical Abstract.